# SANKUR KUNDU


Machine Learning Engineer • Software Engineer • Data Scientist

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Engineer who ships systems end to end. Co-founder and Director of Technology at Gigzen, where I built both products alone: Buzz Buzz, a 13K-line Android app on Postgres row-level security, and Populace, a testing platform that surfaces the multi-user bugs one developer cannot reach. 15K+ lines of production code, all of it public and running.

## EXPERIENCE

**Freelance — Data Scientist** — *Contract via Upwork* Remote, client in Spain | Jun 2026 – Present

- Engineer Python/Pandas/NumPy/SQL pipelines over multi-GB census, commercial, and STATA .dta sources, converting multi-year raw data into analysis-ready datasets for an international research team.
- Collaborate cross-functionally with researchers and statisticians to build validation and reproducibility checks into every deliverable, shipping statistical modeling inputs and analytical outputs under fixed publication timelines.

**Gigzen Private Limited** — *Co-founder & Director, Technology* Bhubaneswar, India | Apr 2026 – Present

- Co-founded an MCA-registered company targeting the USD 582B gig economy and lead its technical function, owning the products end to end: architecture, implementation, database design, release engineering and compliance.
- Own the infrastructure both products run on and the engineering standards a second contributor works to.

## PRODUCTS

*Gigzen Private Limited — shakhtar-sankur.github.io/gigzen. Both products below are public and linked.*

**Populace** — *Multi-user testing platform* Shakhtar-Sankur/populace

- Designed an agent-based platform where concurrent synthetic users exercise a real app through its own API as real authenticated accounts, surfacing presence, sync, permission and fan-out bugs one developer cannot reach.
- Built an app-agnostic engine with zero runtime dependencies behind a 13-method adapter contract, per-endpoint p50/p95/p99 latency instrumentation that shows how an app degrades as agent count rises, exposing scalability limits before real users hit them, and three independent guards that refuse to run against production; exits non-zero on any failure so it gates CI.
- Made honest reporting a design constraint: an empty adapter method does not count as coverage, so a fresh scaffold reports 2/13 and refuses to run rather than passing while testing nothing. 20/20 self-tests against an in-memory adapter.

**Buzz Buzz** — *Gig-worker platform, Android* Shakhtar-Sankur/buzz-buzz

- Built a 13,235-line React + TypeScript + Capacitor app on Supabase: 17 Postgres tables under 48 row-level-security policies, realtime sync, presence, read receipts, GPS tracking, photo posts and groups.
- Localized into 16 languages including right-to-left Arabic, with currency and language selected automatically by region.
- Debugged a recursive row-level-security policy that silently broke every chat read, fixed it with a SECURITY DEFINER helper, and verified the full stack end to end against a live backend with real accounts.

## SELECTED ML PROJECTS

*All public at github.com/Shakhtar-Sankur — each README states its design targets and what the code measures against them.*

- **ML Serving Platform** — The operational layer around inference: versioned rollout, A/B routing, circuit breaking, per-feature drift detection, caching and metrics. Designed for 1,200 QPS. *Flask, Redis, Triton, Kubernetes, Prometheus, Docker*
- **Edge AI Model Compression** — Compression pipeline for battery-powered edge inference: distillation from a ResNet50 teacher, magnitude pruning, int8 quantization, ONNX export. Measured results in the repo. *PyTorch, ONNX, ONNX Runtime*
- **IoT Predictive Maintenance on GCP** — LSTM failure prediction on a 48-hour forecast horizon with chronological splits, Optuna tuning, FastAPI serving and a GCP deployment path. *TensorFlow, Optuna, FastAPI, Pub/Sub, BigQuery, Vertex AI*
- **DETEK — Data-Leak Detection on AWS** — Pairs a BERT content classifier with an LSTM behavioural model, because either alone produces unusable alerts. CloudTrail and VPC Flow collectors, Terraform infrastructure. *PyTorch, Transformers, Flask, boto3, Terraform*

## SKILLS

**Languages:** Python, TypeScript, SQL, R, STATA

**ML & AI:** Production ML pipelines, model training and evaluation, predictive and statistical modeling, deep learning (LSTM, BERT, DDPM), knowledge distillation, pruning and int8 quantization, ONNX export, anomaly detection, LLM agent orchestration, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

**Systems & Cloud:** System design, distributed systems, CI/CD, unit and integration testing, GCP (BigQuery, Vertex AI, Dataflow, Pub/Sub), AWS (Lambda, S3, EventBridge, CloudTrail), FastAPI, Flask, Node.js, React, Capacitor, PostgreSQL and Supabase (row-level security), Redis, Triton, MLflow, Optuna, Prometheus, Docker, Kubernetes, Terraform

## EDUCATION & CERTIFICATIONS

**Institute of Engineering and Management, Kolkata** — B.Tech, Computer Science & Engineering 2022 – 2026

**Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Computer Networks, Computer Architecture, Machine Learning

**Certifications:** Meta Data Analyst; Google Data Analytics; Google Data Analysis with R; Google Cybersecurity; IBM Cybersecurity Essentials; Microsoft Building AI Cloud Apps with Azure; Google Cloud learning paths (Data Engineer, Cloud Architect, DevOps, Database Engineer); AWS Architecting Solutions, Building Data Lakes, Migrating to AWS